

Some notes on burritos

When we want to take the derivative of a composite function $f(g(x))$, we use the Chain Rule:

$$\frac{d}{dx} [f(g(x))] = f'(g(x)) \cdot g'(x)$$

Composite functions are like **burritos**: they are made of a tortilla wrapped around a filling. In the setup $f(g(x))$, I am saying that the tortilla is $f(\sim)$ and the filling is $g(x)$.

I am saying this in kind of a weird way, and for a specific reason. I am saying that the tortilla is $f(\sim)$, rather than $f(x)$, because I am emphasizing that whatever the input is for the function $f(\sim)$, it is in some way **more complicated** than just plain x .

In particular, when we write down a composite function $f(g(x))$, exactly what we are doing is:

- starting with a plain-old-boring $f(x)$,
- choosing some other function $g(x)$, and most importantly
- **replacing** the x 's in plain-old-boring $f(x)$ by $g(x)$.

And thus we have created a burrito, where we have taken a tortilla and put in some filling $g(x)$.

How to recognize burritos

A very common question that people have when they are learning the chain rule for the first time is, how do I know what is a burrito and what is not, and how do I tell what is the tortilla and what is the filling?

One way to approach this question is to think about a library of possible plain-old-boring functions to start with. I will in particular suggest looking at the list of derivative rules that are about specific functions:

☐ power function rule: $\frac{d}{dx}[x^n] = nx^{n-1}$

☐ exponential function rule: $\frac{d}{dx}[a^x] = a^x \cdot \ln(a)$

trigonometric function rules:

☐ $\frac{d}{dx}[\sin(\boxed{x})] = \cos(\boxed{x})$

☐ $\frac{d}{dx}[\cos(x)] = -\sin(x)$

☐ $\frac{d}{dx}[\tan(x)] = \sec^2(x)$

☐ logarithmic function rule: $\frac{d}{dx}[\ln(x)] = \frac{1}{x}$

Before moving on, take a minute and highlight for yourself the **variable** in each of these derivative rules. Do this on both sides of the equals sign. I have done this for the one about the sine function as an example.

How to recognize burritos, continued

So here's the deal: in the context of the chain rule, you can spot a burrito if you see one of these functions, but instead of a just-plain- x (or, y'know, just-plain t or z or whatever letter the variable is), you see another function.

In other words, you are **expecting** to see a just-plain- x inside some function, but instead you see **something more complicated**.

Here are some examples, with the filling highlighted:

$$\bullet \left(4z^3 + 77 \right)^{17} \quad \bullet \sin(7t) \quad \bullet \cos(\ln(q)) \quad \bullet e^{2x+6}$$

And here are the corresponding tortillas:

$$\bullet (\sim)^{17} \quad \bullet \sin(\sim) \quad \bullet \cos(\sim) \quad \bullet e^{\sim}$$

Using the chain rule

(Placeholder. More to come here!)